Alfie R. Pelicano

Full-Stack Developer

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Professional Summary

Research Specialist and Geodetic Software Developer with 16+ years in scientific computing and spatial data. I work at the intersection of earth science and software engineering, processing GNSS and geophysical data for earthquake research and fault monitoring while building the software to automate what shouldn't still be manual.

Specialized in designing headless processing pipelines, distributed task systems, and full-stack applications. I build to make impossible or tedious work tractable, prioritizing minimalist, low-cognitive-load architectures over complex stacks.

Currently developing and maintaining flagship earth science systems (MOVE Faults, EIL Hazard Suite), civic tech platforms (YAKAP-Link), and B2B marketplaces. My development process is specification-driven and test-backed: defining strict data contracts and system behaviors upfront, and ensuring flawless execution through rigorous test coverage.

Core competencies: GNSS Processing, Spatial Geohazards, Full-Stack Web Development, Agentic AI Workflows, Distributed Task Queues (Celery/Redis), and Relational Databases (PostgreSQL).

Technical Skills

Frontend	Next.js 14 (App Router), React 19 (Server Components), Astro (Static Site Generation), TypeScript, Vanilla JavaScript, Tailwind CSS, shadcn/ui
Backend	Python / FastAPI, Node.js / Express, Celery (Distributed Task Queues), Redis, Docker, Agentic AI Workflows (LangGraph / Ollama), Prisma ORM, NextAuth.js
Database & Infra	PostgreSQL (Advanced SQL), Neon (Serverless), Supabase, Row Level Security (RLS)
Python & Scientific	Python 3.11+, Click CLI, Rich Terminal, Pandas, NumPy, SciPy, Shell Scripting, Fortran (Legacy Scientific Code)
Domain Expertise	GNSS Processing & Field Ops, Spatial Geohazards (Landslide / Seismic), Scientific Computing & HPC, Heuristic Engines, Constraint Solving
Testing & Quality	Spec-Driven Architecture & TDD, Playwright (E2E), Vitest (Unit), Pytest, Git Workflows
Tools & Workflows	Agentic Workflow Design & Orchestration, LaTeX / Beamer, ADR-driven Architecture, Linux / Bash

Production Applications

2024–Present **PipetGo – B2B Laboratory Testing Marketplace**, *Next.js 14 · TypeScript · Prisma · NextAuth · PostgreSQL*

Production MVP connecting businesses with ISO 17025 accredited testing labs.

- Implements multi-tiered pricing modes (QUOTE_REQUIRED, FIXED, HYBRID) and multi-role authorization workflows.
- Undergoing User Acceptance Testing with initial partner organizations before pilot deployment.

Repository: https://github.com/alfieprojectsdev/pipetgo

2024–Present **PHAST – PHIVOLCS Hazard Assessment Tool**, *Vanilla JS · Chrome Extension API · Leaflet.js*

Chrome Extension command center that injects professional GIS controls into standard web maps.

- Dynamically generates formatted, official hazard assessment text directly from map interactions.
- Replaces manual text authoring for hazard assessors.

Repository: https://github.com/alfieprojectsdev/phat

2023–Present **Window Cards Generator**, *Vanilla JS · HTML5 · CSS3 · Print CSS*

Zero-dependency Vanilla JS constraint-solving engine that dynamically generates printable math problem grids.

- Designed for SPED and early-grade classrooms with limited device access/connectivity.

Live Demo: https://alfieprojectsdev.github.io/windowcards

2024 **ArtPortfolio – Statically Generated CMS**, *Astro · Static Site Generation · GitHub Pages*

High-performance Astro CMS built for digital art commissions, optimized for minimal cognitive load and fast asset delivery with zero client-side JS by default. **Repository:** https://github.com/alfieprojectsdev/artportfolio

Active Development Projects

2024–Present **MOVE Faults & EIL Hazard Suite**, *Python · Celery · React · FastAPI · Docker*

- **MOVE Faults:** Flagship GNSS automation monorepo replacing legacy scripts. Features a Celery-backed distributed task queue, real-time NMEA TCP streaming, and an offline React field app.
- **EIL Hazard Suite:** Decoupled geohazard system featuring a headless spatial processing engine (EIL-Calc) for landslide susceptibility scoring and a React visualizer (EIL-Viz) for ground-truth validation.

2024–Present **Civic & Utility Platforms**, *Next.js 14 · TypeScript · PostgreSQL · Prisma / Drizzle*

- **YAKAP-Link:** Civic tech POC with strict transaction constraints and dispensing guard logic to mitigate phantom claims in the PhilHealth medicine program.
- **Homebase:** ADHD-optimized household command center with heuristics for anomaly detection, distributed notification loops, and multi-residence bill tracking.
- **Washboard:** Contactless QR car wash booking system replacing manual pen-and-paper queue tracking with a real-time receptionist dashboard.

2024–Present **Agentic Orchestrators (NightCoder & SciWriter)**, *Python 3.11+ · Ollama · Local LLMs · Shell Scripts*

- **NightCoder:** ADR-to-patch automation engine utilizing local LLMs (deepseek-r1, qwen2.5-coder) to synthesize structured implementation plans and git-style patches.
- **SciWriter:** Config-driven writing pipeline that converts research notes into structured drafts with refinement passes, terminology normalization, and LaTeX/Beamer export.

2024–Present **Drive Archaeologist & SIPAT**, *Python · TypeScript · React · IndexedDB*

- **Drive Archaeologist:** Python CLI for excavating legacy GNSS data. Detects RINEX files, flags MD5 duplicates, and generates safe migration scripts.
- **SIPAT:** Open-source CAPI alternative for field researchers needing offline-first data collection, complex branching logic, and automated audio transcription.

Work Experience

2009–Present **Science Research Specialist**, *Philippine Institute of Volcanology and Seismology (PHIVOLCS)*

- Process GNSS and geophysical data for tectonic monitoring, earthquake research, and fault assessment across the Philippines.
- Design and build headless processing pipelines, distributed task systems, and full-stack applications (e.g., MOVE Faults, EIL Hazard Suite, PHAST) to automate decades of legacy scientific workflows.
- Participate in GNSS field surveys and network maintenance, supporting station integrity and data continuity.
- Contribute as co-author to peer-reviewed publications on crustal deformation and geodetic methods.
- Transitioned technical skills to modern web development, building production applications that apply scientific rigor (TDD, strict data integrity) to software engineering.

2005–2009 **Software Test Engineer**, *Azeus Systems Philippines*

- Performed functional and regression testing for large-scale web applications, including government contract platforms.
- Wrote detailed manual test cases and reported issues with clear reproduction steps, improving dev handoff quality.
- Used SQL for test setup and validation of back-end logic and edge cases during integration testing.

2004–2005 **Science Information Analyst**, *ESS Manufacturing Company, Inc. (EMCI)*

- Indexed and summarized physics and engineering articles for an international scientific publication database.
- Applied technical reading and abstract writing skills under tight deadlines across multiple disciplines.

2002–2003 **Student Assistant**, *National Institute of Physics – UP Diliman (Theoretical Physics Group)*

- Maintained Linux workstations for basic network access, file management, and local simulations.
- Helped students troubleshoot command-line issues, particularly Fortran compilation and plotting with gnuplot.
- Assisted with recovering and digitizing legacy scientific code from non-machine-readable sources.

Education

1999–2004 **BS Applied Physics, Instrumentation**, *University of the Philippines Diliman*
Eugenio Lopez Jr. Scholar (2002–2004)

1995–1999 **Electronics Technology Track**, *Don Bosco Technical College*

2015 **Master of Geoscience (Partially Completed)**, *Macquarie University*
Completed a semester-length unit on scientific data scripting and visualization, covering UNIX shell, GMT, Terragen, Blender, and Python-based workflows

Select Publications & Research

2023 *Geophysical Research Letters:* Oblique blind faulting, Abra Earthquake

2013 *Journal of Geophysical Research: Solid Earth:* Interseismic deformation along the Philippine Fault

2015 *Journal of Disaster Research:* Plate convergence and GPS motions in Mindanao

2010–2013 *AGU Fall Meeting Abstracts:* Deformation modeling and GPS-based intrusion studies

(Full list available on request or via LinkedIn/Google Scholar profile)